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Effect of Membrane Cholesterol on Calcium Phosphate Formation in Aqueous Suspensions of Anionic Liposomes

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Summary. The present study examined the effect of membrane cholesterol on liposome-mediated calcium phosphate precipitation in metastable aqueous solutions (2.25 mM Ca^{2+} and 1.5 mM inorganic phosphate) at 22°C, pH 7.4 and 240 mOsm. The liposomes were prepared from 7:2:X molar mixtures of phosphatidylcholine, dicetylphosphate, and cholesterol (x = 0, 1, 5, or 9) and contained either 0 or 50 mM encapsulated phosphate. The membranes were made permeable to Ca^{2+} by addition of the cationophore, X-537A. Changes in external Ca^{2+} concentration were used as the principal monitor of the course of precipitation. Without encapsulated phosphate, 7:2:X liposomes (with or without ionophore) induced no precipitation. With 50 mM encapsulated phosphate and in the presence of ionophore, precipitation significantly depended on the cholesterol level in the membrane. At 0 and 10 mole% cholesterol, precipitate developed rapidly both within and outside the liposomes. At 35 and 50 mole% cholesterol, no observable intraliposomal precipitation occurred, and extraliposomal precipitation started only after an induction period of 24 hours. Delayed extraliposomal precipitation also took place in PO_4-containing liposomes without added ionophore. In this latter case, however, cholesterol was essential for this precipitation to occur with the optimum level being around 10 mole%. Suppression of ionophore-mediated intraliposomal precipitation at higher cholesterol levels could be related to the inflexible cholesterol molecules making the membrane more rigid, thereby restricting Ca-ionophore transport. This restriction could be reversed with ethanol. Delayed extraliposomal precipitation in the absence of added ionophore (or at higher cholesterol levels in its presence) could be explained by seeding from low, unobserved levels of intraliposomal precipitate formed during slow, unfacilitated Ca^{2+} leakage into the liposomal interior.

Key words: Calcium phosphates – Liposomes – Cholesterol – Matrix vesicles.

Since their introduction in the 1960s [1], artificial lipid vesicles (liposomes) have been extensively used as experimental models of biological membranes. Liposomes have been especially useful in studying the relation of membrane composition to structure and function. One substance extensively studied in this regard has been cholesterol (Chol), a major component in all biological membranes. In general, it has been found that Chol in liposomes stabilizes membrane structure and inhibits various membrane-centered functions. As examples, the introduction of Chol into phosphatidylcholine (PC) liposomal membranes decreases membrane permeability and association of membranes with cytoplasmic protein [2], stabilizes membrane order [3], and inhibits phosphatidate-mediated calcium traversal of liposome bilayers [4]. Together with phosphatidic acid (PA), phosphatidylserine (PS), and/or phosphatidylinositol (PI) in hemoglobin containing PC liposomes, Chol significantly reduces or almost completely inhibits lipid peroxidation [5].

In several recent publications [6–10] we have reported the use of anionic liposomes composed of 7:2:1 molar mixtures of PC, dicetylphosphate (DCP), and Chol as useful models for examining membrane-mediated calcium phosphate precipitation. In the present article, we report some kinetic data that reveal that the effect of increasing the membrane Chol level in these liposomes is to delay or inhibit calcium phosphate formation. As these liposomes have many features in common with biological matrix vesicles [6–10], the results may have a bearing on understanding the dynamics of matrix vesicle calcification as well as on calcification processes in certain pathological lipidic conditions, for example, coronary and/or vascular diseases where Chol accumulates in atherogenic plaques prior to mineral deposition.

Materials and Methods

All chemicals used for preparing the liposomes and for carrying out the precipitation experiments had a stated purity of 99% or greater and were used without additional purification. PC (egg lecithin), DCP, and Chol were obtained from Avanti Polar Lipids Inc. (Birmingham, AL), Sephadex G-50C from Pharmacia Fine Chemicals (Piscataway, NJ), lasalocid acid (X-537A) from Aldrich Chemical Company (Milwaukee, WI), and PIPES and HEPES buffers from Sigma Chemical Company (St. Louis, MO). Other chemicals used (NaCl, KCl, KH_2PO_4, $CaCl_2$) were of reagent grade quality. CO_2-free, deionized-distilled water was used in all solution preparations. PC, a neutral bilayer-forming phospholipid, and DCP which facilitates formation of aqueous compartments within the liposome, were used as the basic lipid components in all the liposome preparations. Four levels of Chol were examined: 0 mole % (7PC:2DCP), 10 mole

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Certain commercial materials and equipment are identified in this paper to specify the experimental procedure. In no instance does such identification imply recommendation or endorsement by the National Institute of Standards and Technology or that the material and the equipment identified is necessarily the best available for the purpose.

% (7PC:2DCP:1Chol), 35 mole % (7PC:2DCP:5Chol), and 50 mole % (7PC:2DCP:9Chol). All liposomes were prepared as described elsewhere [6, 10] by rotary evaporating to a thin film the lipids from chloroform under vacuum, followed by dispersing the film with mechanical agitation in PIPES-buffered (pH = 7.4) KCl solution containing 0 or 50 mM inorganic phosphate (PO_4). The pH and osmolality of all solutions was adjusted to 7.4 and 240 mOsm, respectively, before use.

In order to obtain liposomes more homogeneous in size and bilayer thickness, suspensions were subjected to a combination of freeze-thawing (FT) and high pressure extrusion (HPE) [11]. After a minimum five cycles of freezing (−80°C, dry ice/ethanol) and thawing (37°C, water bath) the suspensions were subjected to 10 repetitive passes through two stacked 0.4 μm pore polycarbonate filters under 100 psi argon in a HPE apparatus (Sciema, Richmond, B.C., Canada). Liposomes prepared by the FT-HPE technique were then aged for approximately 22 hours at 22°C (temperature at which all precipitation experiments were carried out). Unencapsulated phosphate was removed by gel filtration (Sephadex G-50C gel column, eluted with HEPES-buffered NaCl), and collected fractions richest in liposomes were pooled for the precipitation reactions. Appropriate volumes of PIPES-buffered 50 mM KCl or 50 mM KH_2PO_4 and HEPES-buffered 4.5 mM $CaCl_2$ solutions were added to 2.0 ml fractions of pooled liposome suspensions in order to prepare systems with a final total lipid concentration of 2.5 mM and a concentration of 2.25 mM Ca^{2+} and 0 or 1.5 mM total inorganic phosphate (PO_4) in the external solution phase.

The liposome membranes were made permeable to external Ca^{2+} ions by addition of the cationophore, lasalocid acid (X-537A). The ionophore was added either in chloroform at the beginning of the liposome preparation or injected as a 0.6 mM ethanolic solution after Ca^{2+} addition. An ionophore/lipid molar ratio of 0.003 was used, which, as previously shown [12], was sufficient to generate Ca^{2+} fluxes capable of inducing intraliposomal precipitation in PO_4-containing 7PC:2DCP:1Chol liposomes. Reaction kinetics were monitored by filtering 0.5 ml aliquots of the suspension at regular intervals through 0.22 μm nitrocellulose membranes and measuring filtrate Ca^{2+} by atomic absorption spectrophotometry.

Changes in Ca^{2+} concentration corresponding to precipitation in the external solution phase made metastable with 1.5 mM PO_4 (i.e., extraliposomal precipitation) were calculated by subtracting from the total solution calcium loss (i.e., initial Ca^{2+} minus filtrate Ca^{2+}) that which resulted from Ca-lipid binding and intraliposomal precipitation. The latter values were obtained from control experiments carried out in the absence of external phosphate.

In some experiments, encapsulated inorganic phosphate and total sample phosphate were determined by applying, respectively, the UV spectrophotometric method of Murphy and Riley [13] and the $HClO_4$ digestion method of Rouser et al. [14]. Lipid PO_4 was calculated as the difference between total sample phosphate and that due to encapsulated inorganic phosphate.

Representative samples of the original suspensions of 7:2:1 and 7:2:9 PC-DCP-Chol liposomes, which were negatively stained with 4% aqueous ammonium molybdate [9], were examined in a JEOL 2000FX transmission electron microscope (TEM) operating at 100 KeV.

Results

The presence of Chol *per se* in the liposomal membrane did not affect the stability of the external suspension medium toward unseeded precipitation in 7PC:2DCP:XChol liposome suspensions. In the absence of ionophore and encapsulated phosphate, none of the Chol-containing liposomes induced calcium phosphate precipitation in metastable external solutions for at least 72 hours, as measured by a constancy in the external Ca^{2+} concentration over this period of 2.21 ± 0.05 mM for all preparations. However, when X-537A ionophore was added to the lipid-$CHCl_3$ solution before vesiculation, external solution Ca^{2+} losses occurred quite rapidly in 50 mM PO_4-encapsulated 7PC:2DCP and

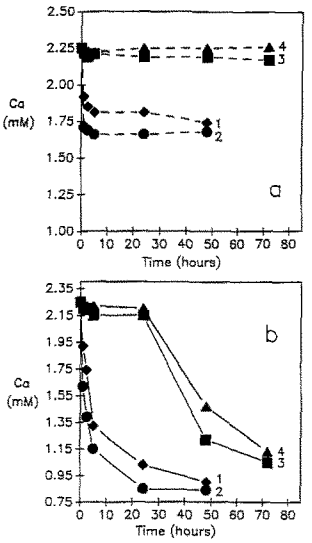

Fig. 1. Kinetics of the change in the external solution Ca^{2+} concentration of X-537A primed liposomes. Ionophore introduced into the $CHCl_3$-lipid solution at the beginning of the liposome preparation procedure. In both panels **(a)** and **(b)**, curve 1 corresponds to 7PC:2DCP liposomes, curve 2 to 7:2:1, curve 3 to 7:2:5, and curve 4 to 7:2:9 PC:DCP:Chol liposomes. All liposomes contained 50 mM encapsulated PO_4, and were suspended in solutions containing **(a)** 2.25 mM Ca^{2+} only or **(b)** 2.25 mM Ca^{2+} and 1.5 mM PO_4. The SD of the plotted values were ≤0.120 mM **(a)** or ≤0.074 mM **(b)**.

7PC:2DCP:1Chol liposome suspensions containing 2.25 mM Ca^{2+} and either 0 (Fig. 1a) or 1.5 mM PO_4 (Fig. 1b) externally. In suspensions without external PO_4, Ca^{2+} losses were due to intraliposomal precipitation [6] and stabilized after 2.5–5 hours at about 0.4 and 0.5 mM for 7:2 and 7:2:1 liposomes, respectively. Additional losses (up to 0.8 mM) in the presence of 1.5 mM PO_4 externally were due to extraliposomal precipitation, and, as reported previously [7], resulted from intraliposomally formed crystals penetrating the liposomal membrane and seeding the external metastable solution. Approximately 80% of these latter Ca^{2+} losses occurred within the first 5 hours. These changes in the external solution Ca^{2+} concentration corresponding to intraliposomal and extraliposomal precipitation are shown separately in Figs. 2a and b and 3a and b. These latter graphs show that both intra- and extraliposomal precipitation occurred somewhat more rapidly initially in liposomes containing 10 mole % Chol than in those that had none.

The data plotted in Figures 1a and 2a show that, in contrast to the 0 and 10 mole % Chol containing liposomes, when PO_4-encapsulated liposomes similarly primed with X-537A but containing 35 or 50 mole % Chol in their membranes were suspended in a PO_4-free external milieu, no significant changes in external solution Ca^{2+} were observed. This apparent absence of intraliposomal precipitation could not be attributed to insufficient PO_4 being initially available for reaction. Incorporating up to 50 mole % Chol into 7PC:2DCP liposomes did not substantially change the amount of inorganic phosphate encapsulated at the time of preparation (Table 1). Likewise, increasing the Chol content from 10 to 50 mole % did not significantly influence the size of the liposomes or the number of membrane lamellae inclosing the entrapped PO_4 solution (Table 1). Moreover, as Figures 1b and 2b show, external solution Ca^{2+} losses still occurred at these higher membrane Chol levels when 1.5 mM PO_4 was added to the external solution, but only after a

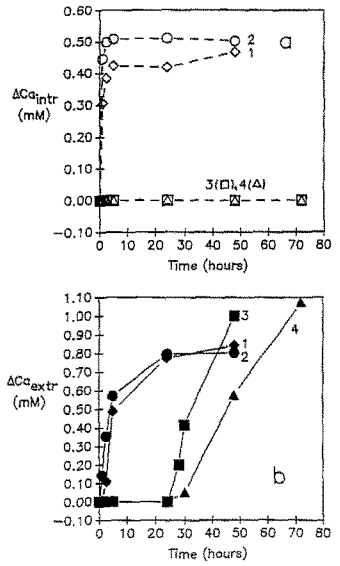

Fig. 2. Changes in the Ca^{2+} losses shown in Figure 1 due to (**a**) intraliposomal (ΔCa_{intr}) and (**b**) extraliposomal (ΔCa_{extr}) precipitation from aqueous suspensions of X-537A primed (1) 7PC:2DCP, (2) 7PC:2DCP:1Chol, (3) 7PC:2DCP:5Chol, and (4) 7PC:2DCP:9Chol liposomes containing 50 mM encapsulated PO_4. External solution contained 2.25 mM Ca^{2+} and 1.5 mM PO_4 initially. The SD of the reported values were <0.110 mM (**a**) or <0.090 mM (**b**).

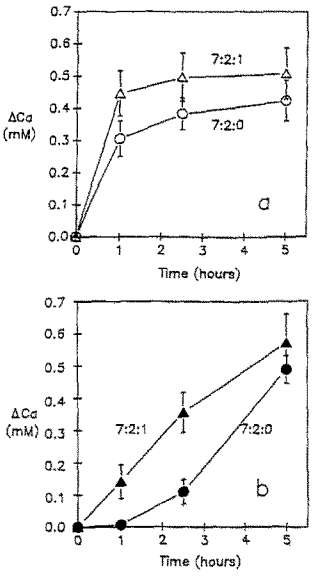

Fig. 3. Changes in intraliposomal (**a**) and extraliposomal (**b**) precipitation as observed during the first 5 hours of precipitation in suspensions of 7PC:2DCP liposomes containing no Chol (7:2:0) and those with 10 mole % Chol (7:2:1). Liposomes were X-537 primed, contained 50 mM encapsulated phosphate, and were suspended in an external solution initially containing 2.25 mM Ca^{2+} and 1.5 mM PO_4. The SD of the plotted values are indicated by bars.

lag period of 24 hours. The total observed Ca^{2+} loss due to extraliposomal precipitation, once the latter began in these high Chol situations, was higher than at lower Chol levels simply because nearly all of the solution Ca^{2+} was available for this precipitation; only small amounts, at most, were diverted to the liposome interior.

Table 1. Size of the liposome, membrane thickness, and the ability to encapsulate phosphate (expressed as the ratio P_{inorg}/P_{lipid}) for 7:2:1 and 7:2:9 PC:DCP:Chol liposome suspensions

	7:2:1	7:2:9
Diameter (nm)[a]	195 (120–380)	220 (110–380)
Membrane lamellarity[b]	4 (2–8)	4 (2–9)
P_{inorg}/P_{lipid}[c,d]	0.206	0.279

[a] TEM data: mean diameter and range ( ) of diameters (75% of population)
[b] TEM data: average number of lamellae per liposome and observed range ( ) of population
[c] P_{lipid} and P_{inorg} measurements performed according to the procedure described in [12]
[d] Values of P_{inorg}/P_{lipid} in 7:2:0 and 7:2:5 PC-DCP-Chol liposomes were 0.182 and 0.196, respectively

As seen from the data in Figure 4, extraliposomal precipitation also occurred when no ionophore was added to the liposomal membrane, as long as PO_4 was encapsulated in the liposomal interior. The necessity for such encapsulated PO_4 was confirmed in ionophore-primed liposome suspensions where no extraliposomal precipitation occurred when the liposomal interior did not contain PO_4 anions. In these latter suspensions, the external Ca^{2+} concentrations were constant at 2.16 ± 0.06 mM over at least 72 hours in the presence of 1.5 mM PO_4 externally. However, as the data in Figure 4 also show, the onset of this X-537A independent, encapsulated PO_4-dependent extraliposomal precipitation was delayed for at least 24 hours in all cases. In addition, the data show that Chol was essential for initiating this precipitation with the optimum level being at around 10 mole % (curve 2, Fig. 4). Higher Chol levels (Fig. 4, curves 3 and 4) slowed but did not totally inhibit this extraliposomal precipitation.

The encapsulated PO_4 leakage data presented in Table 2 show that in the absence of added ionophore, the extent of leakage from Chol-containing liposomes correlated with the amount of extraliposomal precipitate formed after 72 hours (Fig. 4); that is, maximum PO_4 leakage occurred in 10 mole % Chol liposomes, with less seepage and lower precipitate yields occurring in suspensions with higher membrane Chol content. The extent of phosphate leakage also showed the same dependency on the membrane Chol content in 7PC:2PA:XChol, 7PC:2PS:XChol and 7PC:2PI:XChol (for X = 0, 1, 5, and 9) liposome suspensions (Skrtic and Eanes, unpublished results), indicating that this is a general Chol effect in anionic PC liposomes and not the result of a specific interaction with DCP. Surprisingly, lower PO_4 leakage was observed in 7:2:0 than in 7:2:1 liposomes, but as observed with higher Chol-containing systems, slower precipitation kinetics accompanied this reduced leakage (Fig. 3).

In contrast to the above results where X-537A was added to the chloroform-lipid solution at the beginning of the liposome preparation procedure, completely different precipitation behavior was observed when the ionophore was added as an ethanolic solution to already formed liposomes. In this latter case, practically no resistance to X-537A-mediated Ca^{2+} penetration through the membrane could be seen (Fig. 5), even in the presence of 50 mole % Chol. Separately plotting, in this case, the external Ca^{2+} losses due to intraliposomal and extraliposomal precipitation (Fig. 6) also showed no appreciable change in precipitation rate with increased Chol level in the membrane. The slight differences in final intraliposomal precipitation yields in the Chol-containing liposomes were balanced by the extraliposomal Ca^{2+} uptake,

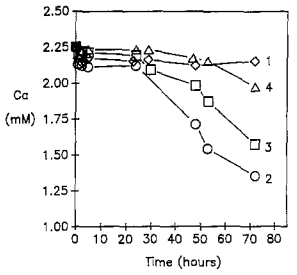

Fig. 4. Changes in the external solution Ca^{2+} concentration as a function of time for suspensions of 7PC:2DCP:XChol (X = 0, 1, 5, or 9—labeled 1, 2, 3, and 4, respectively) liposomes without ionophore, but containing 50 mM encapsulated PO_4. The external solution initially contained 2.25 mM Ca^{2+} and 1.5 mM PO_4. The SD of the reported values were <0.065 mM.

Table 2. Phosphate leakage[a] from 7PC:2DCP:XChol (X = 0, 1, 5, or 9) liposomes containing 50 mM encapsulated PO_4 and no ionophore

Time (hours)	% PO_4 loss 7:2:0	7:2:1	7:2:5	7:2:9
1	3.4 ± 2.8	—	—	—
3	2.3 ± 1.0	11.0 ± 2.5	0	0
24	2.8 ± 2.0	7.5 ± 2.0	3.4 ± 1.0	5.3 ± 2.9
48	2.5 ± 1.4	35.7 ± 11.7	3.9 ± 0.8	4.3 ± 1.3
72	1.7 ± 0.8	35.4 ± 9.7	6.8 ± 2.2	7.9 ± 4.4

Liposomes were suspended in an external solution containing 2.25 mM Ca^{2+}

[a] Leakage measurements performed as described in [6]

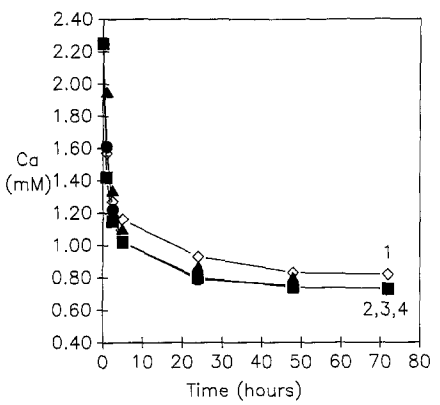

Fig. 5. Comparison of time-dependent changes in the external solution Ca^{2+} concentration for (1) 7PC:2DCP (◇), (2) 7PC:2DCP:1Chol (●), (3) 7PC:2DCP:5Chol (■), and (4) 7PC:2DCP:9Chol (▲) liposome suspensions, in which Ca-liposome interactions were initiated by injecting an ethanolic solution of X-537A. Liposomes contained 50 mM encapsulated phosphate and the external solution contained 2.25 mM Ca^{2+} and 1.5 mM PO_4. The SDs of the plotted values were <0.065 mM.

so that total external solution Ca^{2+} losses in all the Chol systems were stabilized at about 1.5 mM. Only in the Chol-free liposomes was the lower intraliposomal yield not fully compensated by a higher extraliposomal value. As the data in Figure 7 show, Chol-dependent differences in delayed precipitation kinetics also disappeared when ethanol only (i.e., no ionophore) was added to ionophore-free, PO_4-encapsulated liposome suspensions.

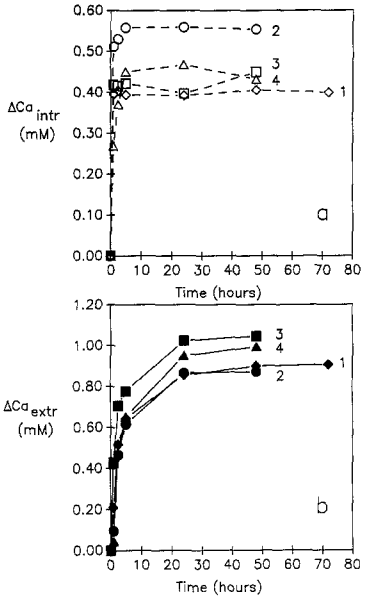

Fig. 6. Changes in the external solution Ca^{2+} losses shown in Figure 5 due to (a) intraliposomal (ΔCa_{intr}) and (b) extraliposomal (ΔCa_{extr}) precipitation. The SDs of the plotted values were <0.105 mM (for intraliposomal precipitation) and <0.060 mM (for extraliposomal precipitation).

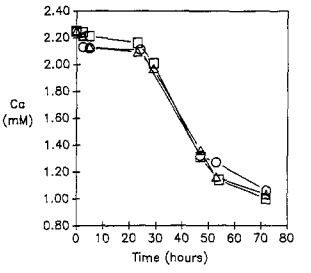

Fig. 7. Kinetic changes in external solution Ca^{2+} concentration as observed when ethanol only (no X-537A) was added to 7PC:2DCP:XChol [X = 1 (○), 5 (□), or 9 (△)] liposomes containing 50 mM encapsulated phosphate. Liposomes were suspended in an external solution containing 2.25 mM Ca^{2+} and 1.5 mM PO_4. Single experiment data.

Discussion

In addition to extrinsic factors such as suspension temperature and supersaturation, and physical factors such as the size distribution of liposomes [12], the kinetics of precipitation from aqueous suspensions of liposomes are strongly affected by several factors that depend directly on the composition of the lipid envelope itself; for example, the nature, strength, concentration, and distribution of the charged ligands on the membrane surface [10]. Results presented in this article show that neutral membrane components such as Chol, which readily incorporate into PC-DCP membranes, can also strongly influence liposome-mediated calcium phosphate precipitation in ethanol-free suspensions; this influence, however, may be largely indirect. For example, there was no evidence that membrane-incorporated Chol *per se* induced precipitation in the external solution phase when the latter was made metastable in Ca^{2+} and PO_4 ions. On the other hand, introducing 10 mole % Chol in PO_4-encapsulated

PC-DCP liposomes resulted in more rapid ionophore-mediated intra- and extraliposomal precipitate development (Fig. 3). Johnson [15] found that the same level of Chol sharply decreased the bilayer thickness of PC:PA liposomes. Perhaps a similar bilayer thinning in PC:DCP liposomes facilitated the effectiveness of the X-537A-Ca transport process. In contrast, 35 and 50 mole % Chol in the liposomal membrane suppressed intraliposomal precipitation. At such high Chol concentrations the membrane-thinning effect is reversed and bilayer thickness increases [15]. In addition, high Chol levels increase the structural ordering of the other lipid constituents [16]. Therefore, high concentrations of the structurally inflexible Chol molecule possibly block X-537A-Ca transport by making the membrane more rigid, thereby restricting transmembrane ionophore mobility.

However, the delayed occurrence of extraliposomal precipitation in those liposome suspensions where ionophore-mediated intraliposomal precipitation was apparently suppressed by a high membrane Chol content is more difficult to explain. The similar delayed appearance of extraliposomal precipitation in Chol liposome suspensions free of any added ionophore appears to rule out Chol-immobilized Ca-X-537A surface complexes on the liposomal exterior as possible seeding sites. On the other hand, the observation that such "spontaneous" precipitation did not occur unless PO_4 was encapsulated suggest that this PO_4 source was involved in the precipitation. The correlation of spontaneous extraliposomal precipitation with the leakage of this PO_4 suggests that local external accumulations at the sites of leakage may be responsible. However, this leakage (Table 2) appears to be so slow that escaping PO_4 would probably diffuse into the bulk of the external solution before an appreciable buildup necessary to initiate precipitation could occur. Another possibility is that if PO_4 can leak out of the liposome interior, Ca^{2+} can enter and react with the interior PO_4 without the assistance of ionophores. As Ca^{2+} is a smaller ion than PO_4 as well as being cationic (the anionic membrane barrier would attract it but repel PO_4), Ca^{2+} should more readily diffuse across the membrane than PO_4. If this residual Ca^{2+} buildup inside the liposome occurs more rapidly than the decrease in resident PO_4 levels, intraliposomal supersaturation to precipitable levels could occur. Therefore, even though no observable Ca^{2+} accumulations were found, it is possible that some intraliposomal precipitation took place and that the penetration of these few crystals to the external liposomal surface where they could act as seeding substrates may explain the "spontaneous" precipitation. The paucity of these crystals due to the slowness of the diffusion process may account for the lengthy induction that is observed before the onset of the spontaneous precipitation (Fig. 2b).

However, the rapidity by which intraliposomal precipitates can form and subsequently seed extraliposomal precipitation in low (0–10 mole %) Chol liposome suspensions suggests that when ionophores are operative, their efficiency in transporting Ca^{2+} is so much greater as to render nonionophoretic diffusion processes unimportant in their presence. Only when the ionophore is absent or cannot function, as in the case of high Chol (>35 mole %) membranes, can these latter, slower processes contribute significantly to extraliposomal precipitate formation.

Once established, extraliposomal precipitation was found to expand rapidly throughout the external solution phase in all suspensions (Fig. 2b). This latter result suggests that Chol does not interfere with extraliposomal precipitation once it begins, a behavior that is in marked contrast to such acidic lipid substances as PS and PA which were found [8, 10] to appreciably retard extraliposomal mineral growth by inducing the adherence of liposomal membranes to the surfaces of forming crystals, thereby restricting the availability of these surfaces as growth sites [10]. Instead, as stated above, Chol (at higher membrane concentrations) appears to delay precipitation primarily by interfering with transmembrane Ca^{2+} and PO_4 movement.

The absence of Chol inhibition of liposomal precipitation when X-537A was added after liposome formation (Fig. 5) was probably due more to the manner in which the ionophore was introduced than to the time of addition, as similar results on "spontaneous" precipitation were observed when the ethanol carrier alone was added (Fig. 7). The most probable explanation for this negating effect of ethanol on the Chol results is that either the ethanol interfered with Chol's ability to organize other membrane constituents into rigid, ionophore-immobilizing arrays or it established regions of such high fluidity and permeability in the membrane that the stiffening effect of Chol was not a significant deterrent to either passive or facilitated ion movements through these regions.

The membrane envelope of cartilage matrix vesicles, the nidi for calcification in this tissue [17–19], is markedly enriched in Chol (as high as 82% of the phospholipid content [20]) compared with the chondrocytic plasma membrane from which it was derived [20, 21]. The results of this study indicate that such elevated Chol concentrations probably do not have a direct seeding effect on mineral formation in these extracellular structures. On the other hand, it is tempting to conclude by analogy to our liposome results that this elevation of Chol in the vesicle membrane may possibly adversely affect Ca^{2+} transport into the vesicle interior, especially as passive rather than energy-demanding active transport pathways are largely responsible for the inward flux of this ion necessary to initiate and sustain this mineral formation [22–25]. However, the extension of our results to the *in vivo* situation is complicated by the finding that, in addition to phospholipids such as PC, the matrix vesicle membrane contains a diverse mixture of other lipidic substances, especially free fatty acids, triacylglycerols, and glycolipids [20, 21]. The effect these other lipids have on Ca^{2+} transport and matrix vesicle mineralization as well as on the role Chol plays in these processes remains to be established.

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